

Kia Ghods

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Last updated April 27, 2026

Education

Princeton University

BSE in Computer Science

2023 – 2027

- Minors in Mathematics, Statistics & Machine Learning; GPA: 3.93/4.0
- Full list of major coursework: <https://kiaghods.com/academics>

Honors & Awards

- Honors: Shapiro Prize for Academic Excellence (top 3% of Princeton underclassmen), Tau Beta Pi, Neo Scholar Finalist, ICLR 2026 Gen² Workshop Student Travel Grant
- Programs: Princeton Machine Learning Theory Summer School ([\[website\]](#), Aug 2026)

Research Affiliations

MinRegret Lab, Princeton Computer Science (w/ Prof. Elad Hazan)

Sep 2025 – Present

- Focus on language modeling, robotic control, and alignment
- Developing fully open-source language model FlashSTU 2.0 and small models as probes for large-model dynamics [\[GitHub\]](#)

Princeton Language & Intelligence (PLI) (w/ Profs. Cohen & Griffiths)

Jan 2024 – Jan 2026

- Computational cognitive science of LLMs and VLMs; visual reasoning and in-context compression

Epidemiological Modeling Lab, FAU Mathematics (w/ Prof. Necibe Tuncer)

2021 – 2024

- Mathematical modeling of within-host HIV dynamics; identifiability analysis of models for AIDS clinical trial data

Publications & Pre-prints

*Equal contribution.

- [1] **Kia Ghods*** and Jonathan Liu*. Continuous diffusion transformers for designing synthetic regulatory elements. In *Generative AI in Genomics (Gen²) Workshop at The International Conference on Learning Representations (ICLR)*, 2026. arXiv:2603.10885.
- [2] **Kia Ghods***, Nicholas Budny*, Declan I. Campbell*, Raja Marjeh, Amogh Joshi, Sreejan Kumar, Jonathan D. Cohen, Taylor W. Webb, and Thomas L. Griffiths. Visual serial processing deficits explain divergences in human and VLM reasoning. 2025. Under review. arXiv:2509.25142.
- [3] Declan I. Campbell, Sunayana Rane, Tyler Giallanza, Nicolò De Sabbata, **Kia Ghods**, Amogh Joshi, Alexander Ku, Steven M. Frankland, Thomas L. Griffiths, Jonathan D. Cohen, and Taylor W. Webb. Understanding the Limits of Vision Language Models Through the Lens of the Binding Problem. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2411.00238.
- [4] Necibe Tuncer, **Kia Ghods**, Vivek Sreejithkumar, Adin Garbowit, Mark Zagha, and Maia Martcheva. Validation of a Multi-Strain HIV Within-Host Model with AIDS Clinical Studies. *Mathematics*, 12(16):2583, 2024. doi:10.3390/math12162583.
- [5] Vivek Sreejithkumar, **Kia Ghods**, Tharusha Bandara, Maia Martcheva, and Necibe Tuncer. Modeling the interplay between albumin-globulin metabolism and HIV infection. *Mathematical Biosciences and Engineering*, 20(11):19527–19552, 2023. doi:10.3934/mbe.2023865.

Selected Projects

micrograd-rust

- Built a scalar reverse-mode autodiff engine in Rust (dynamic graph, topo-sorted backward pass, gradient accumulation) and used it to train small neural nets end-to-end; 13–75x faster than Python, scales to 1M+ params where Python stack overflows
- Published as a Python package via PyO3 bindings (`pip install micrograd-rust`); drop-in replacement for Karpathy's micrograd backed by Rust; [\[GitHub\]](#) [\[PyPI\]](#)

[Re]FlashAttention

- Implemented the FlashAttention algorithm (with multi-query attention) in Triton, achieving memory-efficiency and faster processing by avoiding redundant reads/writes between GPU HBM and SRAM; [\[White Paper\]](#), [\[Poster\]](#), [\[GitHub\]](#)

Propagation of In-Context Scheming in Multi-Agent Systems

- Designed experiments to measure how corrupted instructions spread through networks of language models, demonstrating the vulnerability of smaller models to influence peers and higher-capability models
- Awarded 2nd at the Anthropic Alignment Hackathon (2025)

SearchDestroy

- Developed a robust multi-drone search algorithm (DARP + heuristics) to handle drone loss and online path re-computation; created a physics simulation and visualization in AirSim; [\[GitHub\]](#)
- Awarded 2nd at AGI House Robotics Hackathon (2024)

Industry Experience

Astralis

Machine Learning Engineer

Dec 2024 – Feb 2025

- Engineer at early-stage startup using geospatial data to optimize real estate development and demand forecasting
- Built intelligent document processing pipelines (RAG) using Python, LLMs, and ML clustering with automated metrics and async workflows for scalable content organization

Hillspire

Machine Learning Engineer

May 2024 – Dec 2024

- Engineer at a startup at the intersection of consumer social and ML, supported by Sebastian Thrun and Hillspire
- End-to-end development of iOS applications, recommendation algorithms, generative AI, and agentic AI systems

Community

Association for Computing Machinery [\[website\]](#)

Princeton

President, former Outreach Chair

2024 – Present

- Leading Princeton's 600+ member chapter as President, launching initiatives like the “[Princeton Builds](#)” founder program while overseeing 10+ annual events, including the university's largest CS competition ([COSCON](#))
- Organized Natural Language Processing, Computer Vision, and Systems weekly reading groups

Gradient [\[website\]](#)

Admin, Co-Founder

2025 – Present

- Founded Gradient, a 501(c)(3) non-profit incubator for Princeton-originated frontier ventures

AI@Princeton [\[website\]](#)

Princeton

Officer, AI TigerTrek '25 Lead

2024 – 2025

- Led the end-to-end organization of a [selective Silicon Valley immersion](#) for 16 students, securing \$20k+ in funding and sourcing the itinerary with AI leaders from OpenAI, Anthropic, Sequoia Capital, etc.

Teaching Experience

COS 484	Natural Language Processing, Princeton	<i>S26</i>
COS 226	Data Structures & Algorithms, Princeton	<i>S25, F25, S26</i>
MAT 217	Honors Linear Algebra, Princeton	<i>S25</i>
MAC 2312	Calculus II w/ Analytic Geometry, FAU	<i>2021–2022</i>

Skills & Hobbies

Programming Languages	Python, C/C++, Java, Rust, ARM Assembly (AArch64), MATLAB, $\LaTeX$
AI/ML Frameworks	PyTorch, TensorFlow, scikit-learn, Triton
Tools & Systems	Linux, Git, Docker, Kubernetes, PostgreSQL, React
Spoken Languages	English, Persian, Spanish
Hobbies	basketball & strength training; classic film; reading literature